



MCATalyst MCAT Equations

- Shameless plug for my course which I think will REALLY help you, coming soon!:
 - Suck at math and equations and need tons of practice? Click here for my [MCAT Math and Equations Workshop](#)
- [Other Resources, Cheat Sheets, and Tutoring Information](#)
- [Buy me a Coffee/Donate](#)
- [MCAT Mastery App](#)- free app to help you memorize equations, units, SI prefixes, +more!
- [Copy of this document](#) with color coordination so you can prioritize equations

Constants to memorize:

- Avogadro's: 6×10^{23}
- Plank's: 6.6×10^{-34} (although this is often given)
- Gravity: 10 m/s^2
- Speed of Light: $3 \times 10^8 \text{ m/s}$ (this is also often given)
- Fundamental Charge of an electron (and proton): $e_F = \mp 1.6 \times 10^{-19} \text{ C}$ (Coulombs) (this is also often given)
- At STP, 1 mol gas = 22.4L

Standard Unit Prefixes to Memorize:

- Nano (n): 10^{-9} ; 1 nm = 1×10^{-9} meters
- Micro (μ): 10^{-6} , 1 μm = 1×10^{-6} meters
- Milli (m): 10^{-3} , 1 mm = 1×10^{-3} meters (aka .001m)
- Centi (c): 10^{-2} , 1 cm = 1×10^{-2} meters (aka .01m)
- Kilo (k): 10^3 , 1 km = 1×10^3 meters (aka 1000m)
- Mega (M): 10^6 , 1 Megawatt = 1×10^6 Watts

Standard Units

Volume: Liters (L)

Mass: kilograms (kg) in physics, grams (g) in chemistry

Pressure: atmospheres (atm) for everything except fluids (Pascals, Pa)

Temperature: Kelvin (K), NOT Celsius

Amount: moles (mol)

Force: Newtons (N)

Displacement/Distance: meters (m)

Velocity: m/s

Acceleration: m/s^2 (linear) or v^2/r (centripetal/circular)

Work and Energy: Joules

Power: Watts (W)

Period/Time: seconds (s)

Frequency: Hertz (Hz), which = s^{-1}

Intensity: Watts/meters² (W/m^2)

Electric Field: Volts/meter (V/m) or Newtons/Coulomb (N/C)

Magnetic Field: Teslas (T)

Charge: Coulombs (C)

Voltage: Volts (V)

Current: Amps (A), or Coulombs/second (C/s)

Resistance: Ohms (Ω)

Capacitance: Farads (F)

Heat, Enthalpy, and Gibbs Free Energy: Joules (J), just like energy

Entropy: $\text{J}/^\circ\text{K}$

Conversions

- Pressure: 1 atm = 760 torr or mmHg = 100 kPa (rounded!)
 - 1 Pa = 1 N/m^2 (Newtons per meters²)
- Temperature: $^\circ\text{K} = ^\circ\text{C} + 273$
- Parts per million (ppm) = $(\text{amount of solute} / \text{amount of solvent}) * 10^6$
- 1 Joule = eV * $1.6 \times 10^{-19} \text{ J/eV}$
- Area and Volume Conversions: 1 meter will have 10^2 cm, so 1 square m will have $(10^2)^2 \text{ cm}^2$. 1 cubic m will have $(10^2)^3 \text{ cubic cm}$

General ChemistryBasics**Concentration (M)**

$$\text{Molarity (M)} = \text{mol/L}$$

Density, ρ

$$\rho = m/V$$

Molality (m)

$$m = \text{mol}_{\text{solute}} / \text{kg}_{\text{solvent}}$$

Mole Fraction (X)

$$X_A = n_A / n_{\text{total}}$$

Normality (N)

$$N = M * (\# \text{ equivalents})$$

Gram Equivalents

$$m_{\text{eq}} = \text{mass/equivalent weight}$$

$$\text{Eq Weight} = \text{molar mass}/n$$

$$n = \# \text{ equivalents}$$

Degrees of Unsaturation (IHD)

$$\text{IHD} = (2C + 2 + N - H - X) / 2$$

Dilutions

$$M_1 V_1 = M_2 V_2$$

Parts per Million (ppm)

$$\text{ppm} = (\text{amount solute} / \text{amount solvent}) * 10^6$$

Atomic Weight

$$AW = m_1 p_1 + m_2 p_2 + \dots$$

$$p: \text{probability of isotope}$$

Percent Yield

$$\% \text{ Yield} = (\text{Actual}) / (\text{Theoretical}) * 100$$

Equilibrium**Equilibrium Constant, K_{eq}**

$$K_{\text{eq}} = [C]^c [D]^d / [A]^a [B]^b$$

$$\text{For } aA + bB \rightleftharpoons cC + dD$$

$$K_{\text{eq}} = k_{\text{forwards}} / k_{\text{backwards}}$$

$$K_p = P_C^c P_D^d / P_A^a P_B^b$$

$$K_{\text{eq}} \text{ but with Partial Pressures}$$

Reaction Quotient, Q

$$Q = [C]^c [D]^d / [A]^a [B]^b$$

$$\text{Not at equilibrium}$$

$$\text{For } aA + bB \rightleftharpoons cC + dD$$

$$\text{When } Q > K, \text{ reverse rxn favored}$$

$$\text{When } Q < K, \text{ forward rxn favored}$$

$$\text{When } Q = K, \text{ at equilibrium}$$

Gibbs Free Energy Relation to K_{eq}

$$\Delta G^\circ = -RT \ln K_{\text{eq}}, \text{ rearranged:}$$

$$K_{\text{eq}} = e^{-\Delta G^\circ / RT}$$

Kinetics**Arrhenius Equation for Rate Constant, k**

$$k = Ae^{-E_a/RT}$$

Rate Law, rate

$$\text{rate} = k[A]^x[B]^y$$

$$X \text{ and } Y \text{ are orders of A and B}$$

Integrated Rate Laws

- Zeroth: slope = $-k$, Creates straight line on plot where y-axis is $[A]$
- First: slope = $-k$, creates straight line on plot where y-axis is $\ln[A]$
- Second: slope = k , creates straight line on plot where y-axis is $1/[A]$

Thermodynamics**Internal Energy of a Molecule, U**

$$\Delta U = Q - W$$

Work done on/by Gas, W

$$W = -P\Delta V$$

Heat for a ΔT (no phase change)

$$q = mC\Delta T, C: \text{specific heat}$$

Heat of Phase Change

$$q = mL, L: \text{latent heat}$$

Gibbs Free Energy, ΔG

$$\Delta G = \Delta H - T\Delta S$$

Hess Law of Summation

$$\Delta H_{\text{reaction}} = \sum \Delta H_f^\circ \text{ products} - \sum \Delta H_f^\circ \text{ reactants}$$

$$\Delta H_{\text{reaction}} = \sum \Delta H_{\text{bonds broken}} - \sum \Delta H_{\text{bonds formed}}$$

$$\Delta S_{\text{reaction}} = \sum \Delta S_f^\circ \text{ products} - \sum \Delta S_f^\circ \text{ reactants}$$

$$\Delta G_{\text{reaction}} = \sum G_f^\circ \text{ products} - \sum G_f^\circ \text{ reactants}$$

Enthalpy, H

$$\Delta H_{\text{rxn}} = Q/n, n: \text{the number of moles}$$

Entropy, S

$$S = Q/T$$

Thermal Expansion, ΔL

$$\Delta L = \alpha L \Delta T, \alpha: \text{a constant, } L: \text{length}$$

**Gases****Ideal Gas Law**

$$PV = nRT$$

Combined Gas Law

$$P_1V_1/n_1T_1 = P_2V_2/n_2T_2$$

Cancel out constant variables

Dalton's Law of Partial Pressures

$$P_A = X_A P_{\text{total}}$$

$$P_{\text{total}} = P_A + P_B + P_C + \dots$$

Raoult's Law of Vapor Pressure

$$P_A = X_A P_A^{\circ}, P_A^{\circ}: \text{Vapor Pressure}$$

Henry's Law (Solubility of Gas)

$$C_A = kP_A, k: \text{constant for the gas}$$

C: Concentration

P_A : Partial Pressure

$$\uparrow P \propto \uparrow C$$

Graham's Law of Effusion

$$\text{rate}_1/\text{rate}_2 = \sqrt{(m_2/m_1)}$$

Internal Energy of a Gas, U_{gas}

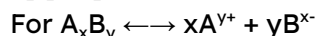
$$U_{\text{gas}} = 3/2 nRT$$

Kinetic Energy of a Gas

$$KE = \frac{1}{2}mv^2 = \frac{3}{2}kT, k \text{ is a constant}$$

Solubility**Solubility Constant**

$$K_{\text{sp}} = [A^{y+}]^x [B^{x-}]^y$$

**Osmotic Pressure, Π**

$$\Pi = iMRT, i: \text{van't Hoff}$$

Boiling Point Elevation

$$\Delta T = iK_b m$$

m : molality, K_b : a constant

Freezing Point Depression/Decrease

$$\Delta T = iK_f m$$

m : molality, K_f : a constant

Retention Factor, R_f

$$R_f = d_{\text{compound}} / d_{\text{solvent}}, d: \text{distance}$$

Acids and Bases**pH**

$$\text{pH} = -\log[H^+] \text{ or } 10^{-\text{pH}} = [H^+]$$

pOH

$$\text{pOH} = -\log[OH^-] \text{ or } 10^{-\text{pOH}} = [OH^-]$$

 pK_a and K_a

$$pK_a = -\log(K_a) \text{ or } 10^{-pK_a} = K_a$$

 pK_b and K_b

$$pK_b = -\log(K_b) \text{ or } 10^{-pK_b} = K_b$$

Ionization Constant of Water, K_w

$$K_a K_b = K_w = 10^{-14}$$

$$\text{pH} + \text{pOH} = 14$$

$$pK_a + pK_b = 14$$

Henderson-Hasselbalch Eq (for Buffers)

$$\text{pH} = pK_a + \log([\text{conj. base}]/[\text{acid}])$$

$$\text{pOH} = pK_b + \log([\text{conj. acid}]/[\text{base}])$$

pI of Amino Acids

$$pI = (pK_{a1} + pK_{a2}) / 2$$

$$\text{Acidic: } (pK_{a,\text{carboxyl}} + pK_{a,\text{R-Group}}) / 2$$

$$\text{Basic: } (pK_{a,\text{amino}} + pK_{a,\text{R-Group}}) / 2$$

$$\text{Neutral: } (pK_{a,\text{carboxyl}} + pK_{a,\text{amino}}) / 2$$

Redox Chemistry and Electrochemistry**Gibbs Free Energy in terms of E**

$$\Delta G = -nFE, n \text{ is moles of electrons}$$

Nernst Equation

$$E = E^{\circ} - (RT/nF)(\log Q)$$

Current caused by n electrons

$$I = nF/t, n: \text{mol } e^-$$

Moles of metal deposited, n_{deposit}

$$n_{\text{deposited}} = It/n_{\text{electrons}} F$$

Cell Potential, E_{cell}

$$E_{\text{Cell}} = E_{\text{reduction,cathode}} + E_{\text{oxidation,anode}}$$

or (they're the same thing)

$$E_{\text{Cell}} = E_{\text{reduction,cathode}} - E_{\text{reduction,anode}}$$

PhysicsTrigonometry and Geometry**SOH CAH TOA**

$\sin\theta = \text{Opposite} / \text{Hypotenuse}$

$\cos\theta = \text{Adjacent} / \text{Hypotenuse}$

$\tan\theta = \text{Opposite} / \text{Adjacent}$

Sin/Cosine to Memorize

$$\sin(90^\circ) = 1$$

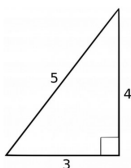
$$\sin(0^\circ) = 0$$

$$\cos(90^\circ) = 0$$

$$\cos(0^\circ) = 1$$

Pythagorean Theorem

$$A^2 + B^2 = C^2$$

3, 4, 5 Triangles

Can be multiples of 3, 4, and 5

Areas

Circle: πr^2

Sphere Surface Area: $4\pi r^2$

Triangle: $\frac{1}{2}\text{base} \times \text{height}$

Kinematics**Displacement, x or d (meters)**

$$d = vt$$

In v vs t plot, d = area under curve

Velocity, v (m/s)

$$v = \Delta d / \Delta t$$

In d vs t plot, v = slope

In a vs t plot, Δv = area under curve

Acceleration, a (m/s²)

$$a = \Delta v / \Delta t$$

In v vs t plot, a = slope

Horizontal displacement

$$\Delta x = v_o \Delta t + \frac{1}{2} a \Delta t^2$$

Final x direction velocity

$$v_{fx}^2 = v_o^2 + 2a\Delta x$$

Work and Energy**Work, W**

$$W = Fd\cos\theta = qEd = qV = \Delta\text{Energy}$$

Power, P

$$P = W/t = Fv = IV$$

Mechanical Advantage

$$MA = F_{\text{assisted}} / F_{\text{unassisted}}$$

Efficiency (can use work OR power)

$$\text{Eff} = P_{\text{output}} / P_{\text{input}}$$

Kinetic Energy

$$KE = \frac{1}{2} mv^2$$

Gravitational Potential Energy

$$U_g = mgh$$

Conservation of Energy

$$U_i + KE_i = U_f + KE_f + W$$

Momentum, p

$$p = mv$$

Force, F

$$F = ma$$

Force of Friction, F_μ

$$F_{\mu} = \mu F_N$$

Forces on an Inclined Plane

$$F_N = mg\cos\theta$$

$$F_{\mu} = \mu mg\cos\theta$$

$$F_g = mg\sin\theta$$

Acceleration Pulley with Two Masses

$$a_{\text{net}} = ((m_1 - m_2)g) / (m_1 + m_2)$$

Center of mass, x_c

$$x_c = (m_1x_1 + m_2x_2 + \dots) / (m_1 + m_2 + \dots)$$

Gravitational Force between Two Objects

$$F_G = G(Mm)/r^2$$

Elastic Force, F_k

$$F_k = -kx, k \text{ is a constant}$$

x: distance from eq. point

Elastic Potential Energy, U_k

$$U_k = \frac{1}{2} kx^2$$

Torque, τ

$$\tau = rF$$

Rotational Equilibrium: $\tau_1 = \tau_2$

Centripetal Acceleration, a_c

$$a_c = v^2/r$$

Centripetal Force, F_c

$$F_c = mv^2/r = ma_c$$

**Fluids****Pressure (Pascal's Law)**

$$P = F/A, A \text{ is surface area}$$

Gauge Pressure (due to depth in fluid), P_G

$$P_G = \rho gh$$

Absolute Pressure (total P in a fluid), P_{abs}

$$P_{abs} = P_{atm} + P_g$$

Buoyant Force, F_b

$$F_b = \rho V_d g = m_d g$$

V_d : displaced fluid volume

m_d : displaced fluid mass

Apparent Force, F_A (F_{net} on obj in fluid)

$$F_A = F_g - F_b, \text{ or:}$$

$$F_b = W_{dry} - W_{wet}$$

Specific Gravity, SG

$$SG = \rho_{object} / \rho_{fluid} \text{ (or } \rho_{substance} / \rho_{reference})$$

$$SG = \% \text{ Volume of Obj Submerged}$$

When comparing two fluids:

$$\rho_{fluid 1} / \rho_{fluid 2} = F_{buoy,1} / F_{buoy,2}$$

For a submerged object:

$$\rho_{object} / \rho_{fluid} = F_g / F_b$$

Volumetric Flow Rate, Q

$$Q = Av, A \text{ is cross-sectional area}$$

Continuity Principle

$$Q_1 = Q_2 \text{ in Ideal Fluids}$$

$$A_1 v_1 = A_2 v_2$$

Bernoulli's Equation

$$P + \frac{1}{2} \rho v^2 + \rho gh = \text{a constant number}$$

$$P_1 + \frac{1}{2} \rho v_1^2 + \rho gh_1 = P_2 + \frac{1}{2} \rho v_2^2 + \rho gh_2$$

Poisuelle's Law

$$\text{Resistance} \propto \text{Radius}^{-4}$$

$$\uparrow 2^* \text{Radius leads to } \downarrow 2^4 * \text{Resist}$$

Atomic/Nuclear Physics**Mass Defect (Nuclear Binding Energy)**

$$E = \Delta mc^2$$

Energy of Electron n Level, E_n

$$E_n = -13.6/n^2, n = \text{energy level}$$

Rydberg Equation (Change in n Level)

$$E_{\Delta n} = hf = R((1/n_f^2) - (1/n_i^2))$$

R = Rydberg constant (will be given)

Number of Half Lives, n

$$N = N_o (\frac{1}{2})^n,$$

n : number of half lives

N : final amount

N_o : is initial amount

Total Time of Half-Life Decay

$$t_{total} = n * t_{1/2}$$

Types of Nuclear Decay

$$\text{Alpha: } {}^A_Z X \rightarrow {}^{A-4}_{Z-2} X + {}^4_2 \text{He (or } {}^4_2 \alpha)$$

$$\text{Beta: } {}^A_Z X \rightarrow {}^A_{Z+1} X + {}^0_{-1} \beta \text{ (or } {}^0_{-1} e)$$

$$\text{Beta}^+: {}^A_Z X \rightarrow {}^A_{Z-1} X + {}^0_1 \beta \text{ (or } {}^0_1 e)$$

$$e^- \text{ Capture: } {}^A_Z X + {}^0_{-1} \beta \rightarrow {}^A_{Z-1} X$$

$$\text{Gamma: } {}^A_Z X \rightarrow {}^A_Z X + \gamma$$

Max # of Electrons in n Energy Shell

$$e^- = 2n^2$$

Photoelectric Effect

$$KE_{\text{electron}} = E_{\text{photon}} - W_{\text{Function}}$$

$$\frac{1}{2} m v_{\text{electron}}^2 = hf_{\text{photon}} - W_{\text{Function}}$$

Circuits**In Series**

I is Constant

$$V_{tot} = V_1 + V_2 + \dots$$

$$R_{tot} = R_1 + R_2 + \dots$$

$$1/C_{tot} = 1/C_1 + 1/C_2 + \dots$$

In Parallel

$$I_{tot} = I_1 + I_2 + \dots$$

V is Constant

$$1/R_{tot} = 1/R_1 + 1/R_2 + \dots$$

$$C_{tot} = C_1 + C_2 + \dots$$

**Electricity and Magnetism****Voltage, V**

$$V = IR = kQ/r = Ed$$

Current, I

$$I = Q/t = nF/t$$

Resistance, R

$$R = \rho L/A, R = 1/\text{Conductivity}$$

Capacitance, C

$$C = \epsilon_0 A/d = Q/V, A: \text{surface area}$$

Dielectric Change in Capacitance, C'

$$C' = kC, k: \text{dielectric constant}$$

Electrical Power, P

$$P = IV, \text{ and since } V = IR:$$

$$P = I^2 R = V^2/R$$

Charge, Q

$$Q = CV = n_{\text{particles}} e_F$$

n : number of particles (not moles)

e_F : is fundamental charge

Force Between Two Charges, F_q

$$F_q = kq_1 q_2 / r^2$$

Electrical Potential Energy, U_q

$$U_q = kqq/r = qV = \frac{1}{2}QV \text{ (and } Q = CV)$$

Electric Field, E

$$E = kQ/r^2 = \Delta V/d = F_E/q$$

Electric Field Force, F_E

$$F_E = qE = q\Delta V/d = ma$$

Magnetic Field Force, F_B

$$F_B = qvB\sin\theta = iLB\sin\theta = mv^2/r$$

$$\uparrow B \propto \uparrow I \text{ (current) and } \downarrow r \text{ (radius)}$$

Dipole Moment, p

$$p = qd$$

Waves**Velocity of a wave, v**

$$v = \lambda f$$

Energy of a Wave

$$E = hf = h(v/\lambda)$$

Period (time) of a Wave

$$T = 1/f$$

Wavenumber (inverse of λ)

$$\text{Wavenumber} = 1/\lambda$$

J to eV Conversion

$$\text{Joules} = eV \cdot e_F$$

e_F : fund. charge of a particle

Electron-Volts, eV

$$eV = N \cdot V, N: \# \text{ electrons}$$

Wave Propagation Phase

$$y(x,t) = A \cdot (\text{wave phase})$$

A : amplitude of wave

Doppler Equation (change in frequency)

$$f' = f(v \mp v_o)/(v \mp v_s)$$

v : velocity of wave

v_o : velocity of observer, v_s : source

+ numerator when moving closer

- denominator when moving closer

Wave Intensity, I

$$I = P/A, A: \text{surface area}$$

P dependent on:

Number of waves/particles

Amplitude of waves

Decibels, dB

$$dB = 10 \log(I/I_o)$$

$$dB + N \cdot 10 \text{ leads to } I \cdot 10^N$$

$$\text{Example: } dB + 20 \text{ leads to } I \cdot 10^2$$

Open Pipe/String Equations (uses 2)

$$f_o = nv/2L, n=1, 2, 3, \dots$$

$$\lambda_o = 2L/n$$

Closed Pipe Equations (uses 4)

$$f_o = nv/4L, n=1, 3, 5, \dots$$

$$\lambda_o = 4L/n$$

Harmonic Frequency, F

$$f_n = nf_o, n: \text{harmonic}, f_o: \text{fund. } f$$

Beat Frequency

$$f_b = |f_1 - f_2|$$

Ultrasound Machines

$$d = vt/2, \text{ thickness} = d_2 - d_1$$

**Optics****Index of Refraction, n**

$$n = c/v$$

c: speed of light in vacuum

v: speed of light in medium

Snell's Law for Refraction

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

Critical Angle, θ_c

$$\sin \theta_c = n_2/n_1$$

Lens Equation

$$1/f = 1/i + 1/o$$

o: obj distance

f: focal length

i: image distance

Radius of Curvature, r

$$r = 2f$$

Magnification, M

$$M = h_i/h_o = -i/o$$

$$\text{Total } M = M_1 * M_2 * \dots$$

Power, P (of a lens)

$$P = 1/f$$

$$\text{Total } P = P_1 + P_2 + \dots$$

Specific Rotation

$$[\alpha] = \alpha_{\text{obs}} / cL$$

$[\alpha]$: specific rotation (degrees)

α_{obs} : observed rotation

c: concentration

L: path length

Biology and Biochemistry**Proteins****Amino Acid Mass**

$$1 \text{ Amino Acid} = 110\text{Da or } 0.11 \text{ kDa}$$

Number of turns in Alpha Helices, n_{turns}

$$n_{\text{turns}} = \text{Total AAs} / 3.6 \text{ AA per turn}$$

Genetics**Number of DNA Copies Produced in PCR**

$$n = n_o 2^N$$

n: number of final copies

n_o : number initial copies

N: number of rounds of replication

Hardy-Weinberg Equilibrium

$$p + q = 1$$

p: dominant allele %

q: recessive allele %

$$p^2 + 2pq + q^2 = 1$$

p^2 : homozygous dominant %

2pq: heterozygous %

q^2 : homozygous recessive %

Physiology**Cardiac Output, SO**

$$CO = HR * SV, \text{ Heart Rate} * \text{Stroke Vol}$$

Vital Capacity, VC

$$VC = IRV + TV + ERV$$

Respiratory Quotient, RQ

$$RQ = \text{CO}_2 \text{ produced} / \text{O}_2 \text{ consumed}$$

Enzymes**Michaelis Menten Equation (Hyperbolic)**

$$V_o = (V_{\text{max}}[S]) / (K_m + [S])$$

$$K_M = [S] \text{ when } V_o = \frac{1}{2} V_{\text{max}}$$

Lineweaver-Burk Equation (Linear)

$$1/V = (K_m/V_{\text{max}})(1/[S]) + 1/V_{\text{max}}$$

$$Y\text{-int} = 1/V_{\text{max}}$$

$$X\text{-int} = -1/K_M$$

$$\text{Slope} = K_m/V_{\text{max}}$$

Turnover Number (Catalytic Const.), k_{cat}

$$k_{\text{cat}} = V_{\text{max}}/[E]$$

Catalytic Efficiency

$$\text{Eff} = k_{\text{cat}}/K_M$$

Probabilities**Probability of Events at the Same Time**

$$p_A * p_B = p_{\text{total}}$$

The probability of A AND B

Probability of Either Events Occuring

$$p_A + p_B - p_{A \text{ and } B} = p_{\text{total}}$$

The probability of A OR B

Number of Orders of Distinct Items

$$n!, n: \text{number of items}$$

Number of Possible Combinations

$$2^n, n: \text{number of items}$$



Equations Organized by Variable

How to Use: This section will be good for learning how to start connecting equations using certain variables. Highly recommend watching my [Puzzle Method](#) video and using this as an easier list of equations to regurgitate.

- Note that I didn't include every single equation (I left out max electrons in an energy shell, as an example), just the ones that have a ton of overlap with a specific variable.

General Chemistry

K_{eq}: $K_{eq} = [C]^c[D]^d/[A]^a[B]^b = k_{forwards}/k_{backwards}$; $\Delta G^\circ = -RT \ln K_{eq}$; $K_{eq} = e^{-\Delta G/RT}$

k: $K_{eq} = k_{forwards}/k_{backwards}$; $k = Ae^{-E_a/RT}$, rate = $k[A]^x[B]^y$

ρ (density): $\rho = m/V$

G: $\Delta G^\circ = -RT \ln K_{eq}$; $\Delta G = \Delta H - T\Delta S$; $\Delta G_{reaction} = \sum GH_f^\circ \text{ products} - \sum GH_f^\circ \text{ reactants}$; $\Delta G = -nFE$

U: $\Delta U = Q - W$; $U_{gas} = 3/2 nRT$

W: $W = -P\Delta V$; $\Delta U = Q - W$

q: $q = mC\Delta T$ (no phase change); $q = mL$ (phase change); $\Delta U = Q - W$; $\Delta H_{rxn} = Q/n$; $S = Q/T$

C (Specific Heat): $q = mC\Delta T$ (no phase change)

L (Latent Heat): $q = mL$ (phase change)

H: $\Delta G = \Delta H - T\Delta S$; $\Delta H_{reaction} = \sum \Delta H_f^\circ \text{ products} - \sum \Delta H_f^\circ \text{ reactants}$; $\Delta H_{reaction} = \sum \Delta H_{bonds \text{ broken}} - \sum \Delta H_{bonds \text{ formed}}$; $\Delta H_{rxn} = Q/n$

S: $\Delta G = \Delta H - T\Delta S$; $S = Q/T$, $\Delta S_{reaction} = \sum \Delta S_f^\circ \text{ products} - \sum \Delta S_f^\circ \text{ reactants}$

P: $PV = nRT$; $P_1V_1/n_1T_1 = P_2V_2/n_2T_2$; $P_A = X_AP_{total}$

V: $PV = nRT$, $P_1V_1/n_1T_1 = P_2V_2/n_2T_2$; $M = \text{mol/L}$

T: $\Delta G^\circ = -RT \ln K_{eq}$; $k = Ae^{-E_a/RT}$; $q = mC\Delta T$ (no phase change); $\Delta G = \Delta H - T\Delta S$; $S = Q/T$; $\Delta L = \alpha L\Delta T$; $PV = nRT$; $P_1V_1/n_1T_1 = P_2V_2/n_2T_2$; $U_{gas} = 3/2 nRT$; $KE = \frac{1}{2}mv^2 = \frac{3}{2}kT$; $\Delta T = iKm$, $\Pi = iMRT$

KE: $KE = \frac{1}{2}mv^2 = \frac{3}{2}kT$

K_{sp}: $K_{sp} = [A^{y+}]^x[B^{x-}]^y$

Π: $\Pi = iMRT$

[H⁺]: $pH = -\log[H^+]$ or $10^{-pH} = [H^+]$; $pH + pOH = 14$; $pH = pK_a + \log([\text{conj. base}]/[\text{acid}])$

[OH⁻]: $pOH = -\log[OH^-]$ or $10^{-pOH} = [OH^-]$; $pH + pOH = 14$; $pOH = pK_b + \log([\text{conj. acid}]/[\text{base}])$

K_a: $pK_a = -\log(K_a)$ or $10^{-pK_a} = K_a$; $K_aK_b = K_w = 10^{-14}$; $pK_a + pK_b = 14$

K_b: $pK_b = -\log(K_b)$ or $10^{-pK_b} = K_b$; $K_aK_b = K_w = 10^{-14}$; $pK_a + pK_b = 14$

pI: $pI = (pK_{a1} + pK_{a2}) / 2$

E (Reductive Potential): $\Delta G = -nFE$; $E_{cell} = E_{reduction, cathode} + E_{oxidation, anode}$ or $E_{cell} = E_{reduction, cathode} - E_{reduction, anode}$

I: $I = nF/t$; $n_{deposited} = It/n_{electrons}F$

F (Faraday's): $I = nF/t$; $n_{deposited} = It/n_{electrons}F$

Physics

d: $d = vt$; $d = vt/2$ for ultrasound; $F_k = -kx$; $U = -\frac{1}{2}kx$; $x_c = (m_1x_1 + m_2x_2 + \dots)/(m_1 + m_2 + \dots)$

v (or c): $v = \Delta d/t$; $a = \Delta v/\Delta t$; $P = Fv$; $KE = \frac{1}{2}mv^2$; $p = mv$; $F_c = mv^2/r$; $a_c = v^2/r$; $P_1 + \frac{1}{2}\rho v_1^2 + \rho gh_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho gh_2$; $Q = Av$; $E = \Delta mc^2$; $v = \lambda f$; $E = hf = h(v/\lambda)$; $f' = f(v \mp v_o)/(v \mp v_s)$; $f_o = nv/(2 \text{ or } 4)L$; $\lambda_o = (2 \text{ or } 4)L/n$; $n = c/v$

α: $\alpha = \Delta v/\Delta t$; $\alpha_{net} = ((m_1 - m_2)g) / (m_1 + m_2)$, $F = ma$, $F_c = ma_c = mv^2/r$

W: $W = Fd \cos \theta = qEd = qV = \Delta \text{Energy}$; $P = W/t = Fv$; $U_i + KE_i = U_f + KE_f + W$; $KE_{electron} = E_{photon} - W_{Function}$

P (Power): $P = W/t = Fv = IV$; $I = P/A$



KE: $KE = \frac{1}{2}mv^2 = \frac{3}{2}kT$; $KE_{\text{electron}} = E_{\text{photon}} - W_{\text{Function}}$; $U_i + KE_i = U_f + KE_f + W$

U (Potential Energy): $U_g = mgh$; $U_q = kqq/r = qV = \frac{1}{2}QV$; $U_i + KE_i = U_f + KE_f + W$; $U_k = -\frac{1}{2}kx$

F (all types of Forces): $F = ma$; $F_\mu = \mu mg \cos \theta$; $F_g = mg \sin \theta$; $F_G = G(Mm)/r^2$; $F_k = -kx$; $F_c = mv^2/r = ma_c$; $F_b = \rho V_d g = m_d g$; $F_A = F_g - F_b$; $P = F/A$; $F = kq_1 q_2 / r^2$; $F_E = qE$; $F_B = qvB \sin \theta = iLB \sin \theta$

p: $p = mv$

τ : $\tau = rF$

P (Pressure): $P = F/A$; $P_G = \rho gh$; $P_{\text{abs}} = P_{\text{atm}} + P_g$; $P_1 + \frac{1}{2}\rho v_1^2 + \rho gh_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho gh_2$

ρ (density): $P_G = \rho gh$; $F_b = \rho V_d g = m_d g$; $SG = \rho_{\text{object}} / \rho_{\text{fluid}}$ (or $\rho_{\text{substance}} / \rho_{\text{reference}}$); $P_1 + \frac{1}{2}\rho v_1^2 + \rho gh_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho gh_2$; $\rho = m/V$

Q (Volumetric Flow Rate): $Q = Av$

Cardiac Output: $CO = HR \cdot SV$

E (Energy): $E = \Delta mc^2$; $E = hf = h(v/\lambda)$; $eV = N \cdot V$; Joules = $eV \cdot e_F$; $KE_{\text{electron}} = E_{\text{photon}} - W_{\text{Function}}$; $E_n = -13.6/n^2$; $E_{\Delta n} = hf = R((1/n_f^2) - (1/n_i^2))$

V: $V = IR$; $P = IV$; In Series: $V_{\text{tot}} = V_1 + V_2 + \dots$; In Parallel: V is Constant; $Q = CV$; $E = \Delta V/d$; $F_E = qE = q\Delta V/d$; $eV = N \cdot V$; $V = kQ/r$

I (Current): $V = IR$; $P = IV$; In Series: I is Constant; In Parallel: $I_{\text{tot}} = I_1 + I_2 + \dots$; $I = Q/t = nF/t$; $F_B = iLB \sin \theta$

R: $V = IR$, $P = V^2/R = I^2 R$ (from $P = IV$ and $V = IR$); In Series: $R_{\text{tot}} = R_1 + R_2 + \dots$; In Parallel: $1/R_{\text{tot}} = 1/R_1 + 1/R_2 + \dots$; $R = \rho L/A$, $R = 1/\text{Conductivity}$;

C (Capacitance): $C = \epsilon_o A/d$; $C' = kC$; $Q = CV$; $U_q = \frac{1}{2}QV$ (and $Q = CV$); In Series: $1/C_{\text{tot}} = 1/C_1 + 1/C_2 + \dots$; In Parallel: $C_{\text{tot}} = C_1 + C_2 + \dots$

Q (Charge): $Q = CV$; $I = Q/t$; $V = kQ/r$; $Q = n_{\text{particles}} e_F$; $F_q = kq_1 q_2 / r^2$; $U_q = kqq/r = qV = \frac{1}{2}QV$; $F_E = qE$; $F_B = qvB \sin \theta$

E (Electric Field): $E = kQ/r^2 = \Delta V/d = F_E/q$

p (dipole moment): $p = qd$

T (Period): $T = 1/f$

λ : $v = \lambda f$; $E = hf = h(v/\lambda)$; Wavenumber = $1/\lambda$; $\lambda_o = (2 \text{ or } 4)L/n$

f: $v = \lambda f$; $E = hf = h(v/\lambda)$; $T = 1/f$; $f' = f(v \mp v_o)/(v \mp v_s)$; $f_o = nv/(2 \text{ or } 4)L$; $f_b = |f_1 - f_2|$

eV: $eV = N \cdot V$; Joules = $eV \cdot e_F$

I (Intensity): $I = P/A$; $d\beta + N \cdot 10$ leads to $I \cdot 10^N$

n (Index of Refraction): $n = c/v$; $n_1 \sin \theta_1 = n_2 \sin \theta_2$; $\sin \theta_c = n_2/n_1$

f (Focal Length): $r = 2f$; $1/f = 1/i + 1/o$; $P = 1/f$

o (Object Distance): $1/f = 1/i + 1/o$; $M = h_i/h_o = -i/o$

i (Image Distance): $1/f = 1/i + 1/o$; $M = h_i/h_o = -i/o$

r (Radius of Curvature): $r = 2f$

P (Power of Lens): $P = 1/f$ (and $r = 2f$); Total $P = P_1 + P_2 + \dots$

M (Magnification of Lens): $M = h_i/h_o = -i/o$; Total $M = M_1 \cdot M_2 \cdot \dots$

α (Specific Rotation): $[\alpha] = \alpha_{\text{obs}} / cL$



Units of Variables

General Chemistry

Concentration

M, Molar

Number of Moles, n

mol, moles

Molar Mass, MM

g/mol, grams per mol

Density, ρ

g/L (grams per liter)

Mass, m

g, grams (in chemistry)

Volume, V

L, Liters

Molality, m

mol/kg

Temperature, T

K, Kelvin

Integrated Rate Law**(Y-Axis)**

0th Order: [A]

1st Order: $\ln[A]$

2nd Order: $1/[A]$

Internal Energy, U

J, Joules

Heat, Q or q

J, Joules

Work, W

J, Joules

Pressure, P

Atm, atmospheres

Gibbs Free Energy, G

J, Joules

Enthalpy, H

J, Joules (often kJ)

Entropy, S

J/K, Joules per Kelvin

Kinetic Energy

J, Joules

Potential Energy, U

J, Joules

Reductive Potential, E

J, Joules

Physics

Displacement, d or x

m, meters

Velocity, v

m/s, meters per second

Acceleration, a

m/s^2 , meters per seconds²

Mass, m

kg, kilograms

Work, W (all units equal)

J, Joules

Nm, Newton meters

Power, P (all units equal)

W, Watts

AV, Amps*Volts

J/s, Joules per Second

Nm/s, Newton meters per second

Kinetic Energy

J, Joules

Potential Energy

J, Joules

Energy, E

J, Joules or eV,
electron-Volts

Force (all units equal)

N, Newtons

$\text{kg}\cdot\text{m/s}^2$, kilogram meters
per seconds²

CV/m, Coulomb Volts per
meter

**Pressure, P (all units
equal)**

Pa, Pascals (in Fluids)

N/m^2 , Newtons per
meters²

Cardiac Output

L/min, Liters per minute

**Current, I (all units
equal)**

A, Amps

C/s, Coulombs per second

Voltage, V

V, Volts

Resistance, R

Ω , Ohms

Capacitance, C

F, Farads

Charge, Q

C, Coulombs

Electric Field, E

V/m, Volts per meter

Magnetic Field, B

T, Teslas

Wavelength, λ

m, meters or nm

commonly for waves

Frequency, f

Hz, Hertz or s^{-1} , inverse
seconds

Period, T

s, seconds

Wavenumber**Intensity**

W/m^2 for Waves

dB sometimes for Sound

Optics Distances (f, o, i)

m, meters

Lens Power, P

D, Diopters